



A Minimalist High-Power Boost Charging Solution Integrated into Electric Drives

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Abstract

The driving capability and charging performance of electric vehicles (EVs) are continuously improving, with high-performance EVs increasing the voltage platform from below 500V to 800V or even 900V. To accommodate existing low-voltage public charging stations, vehicles with high-voltage platforms typically incorporate boost chargers. However, these boost chargers incur additional costs, weight, and spatial requirements. Most mature solutions add a DC-DC boost converter, which results in lower charging power and higher costs. Some new methods leverage the power switching devices and motor inductance within the electric drive motor to form a boost circuit using a three-phase current in-phase control strategy for charging. This approach requires an external inductor

to reduce charging current ripple. Another method avoids the use of an external inductor by employing a two-parallel-one-series topology to minimize current ripple; however, this reduces charging power and increases the risk of torque control during charging. This paper presents an innovative integrated boost charging solution that utilizes a three-phase phase-shift control algorithm. This method eliminates the need for additional external inductors. In addition, intelligent voltage regulation strategies, low-loss motor design, and improved cooling schemes are implemented to address thermal risks associated with motor magnets. Compared to traditional approaches, this method offers advantages in terms of minimal cost increase, reduced space occupation, maximum boost charging power, and safest zero torque output.

Introduction

The swift advancement of electric vehicles (EVs) and escalating charging power requirements have precipitated a consistent elevation in battery voltages. The peak battery voltage for high-performance EVs has incrementally raised from sub-500V to as much as 800V or even 900V, driven by the pursuit of accelerated charging rates, enhanced dynamic capabilities, and weight reduction [1]. Nevertheless, a considerable proportion of existing public charging infrastructure is designed for lower operational voltages: 500V or 750V. These facilities are not readily upgradeable, and such charging networks are anticipated to remain prevalent for an extended duration [2, 3]. As a result, to facilitate compatibility, EVs peak battery voltages surpassing 500V commonly integrate boost converters, enabling adaptation to the lower voltage supply at public charging stations.

EVs equipped with high-voltage systems predominantly utilize separate DC-DC converters for boost charging, incurring additional costs and occupying extra space [1, 4, 5]. Recent years, more integrated solutions

have emerged. For instance, Article [6] introduces an architecture incorporating inductors and relays between the electric drive motor and battery of the EV. This configuration enables the drive motor to serve as a bidirectional charging circuit, capable of both boosting and reducing voltage to accommodate various charging station levels. This approach eliminates the need for a dedicated DC-DC converter, achieving cost reduction and space savings, but it needs significant redesign of the drive motor and the high-voltage network within the EV. Researches [7, 8, 9] have also explored topologies that reuse the motor's inductance and motor inverter for EV boost charging. However, these configurations typically support only AC charging inputs, which are not widely adopted in EVs, for On-board charger is the mainstream AC charging solutions. Back to the integrated solution for DC charging, the drive motor windings often lack sufficient inductance, leading to substantial current ripple, which can result in unsuccessful charging or charging equipment damage. Therefore, leveraging the three-phase windings of the drive motor and inverter to create a boost topology for

DC fast charging often requires an external power inductor to mitigate current ripple, introducing additional components and spatial requirements.

Articles [10, 11] published in 2013 and 2019 proposed utilizing the drive motor's three-phase windings and neutral point, combined with the motor inverter, to form a boost circuit. This arrangement uses the motor neutral point and the high-voltage negative bus to create a charging port, alongside the existing high-voltage positive and negative buses of the electric drive motor, resulting in a "4-port" configuration that realizes the function of boosting the lower voltage of the charging station to the higher voltage of the EV battery. However, these researches remain a preliminary proof-of-concept without addressing motor torque control during boost charging or issues related to motor magnet overheating [12]. Another study [13] suggested employing two three-phase motors to construct a charging topology that adjusts charging voltage and addresses insufficient inductance but is limited to all-wheel-drive vehicles and needs complex architecture. Helpfully, this work provided theoretical support for assessing motor losses when reused as a boost charger. A 2021 study [14] by Hyundai Motor adopted the "4-port" topology introduced in [10, 11], yet found its practical implementation in vehicles to some extent complex and costly. Subsequent publications in 2022 and 2024 [15, 16] using the same topology but introduced novel control algorithms: one involves series connection of one motor phase winding with the other two, while the other employs parallel connection of two phase-windings with interleaved control. These strategies tackle the challenges of inadequate motor inductance causing charging current ripple and excessive motor loss leading to motor magnet temperature overruns. However, [15]'s method fails to ensure zero torque output at all parking positions under boost charging conditions. [16] adjusts motor D-Q current based on stop angles to manage torque, potentially varying the current distribution of two phase-windings. Further analysis shows that different current distribution ratios will lead to inconsistent charging capabilities at different stop angles. Especially when at certain characteristic angles, the charging current is concentrated on one phase-winding; So, at these angles, the charging current will not exceed the continuous current capacity of the motor phase-winding. Neither solution fully exploits the current capacity of the motor's three-phase windings for maximizing charging power.

This study aims to maximize the application value of electric vehicles (EVs) by designing and validating a deep integration of an electric drive motor with boost charging functionality. The optimized topology simplifies the configuration from "4-port" to "3-port," allowing minor modifications to existing electric drive motors to support the boost charging hardware architecture. By improving motor loss optimization and cooling design, combined with software strategies, this study addressed the issue of motor magnet heating. Through the analysis of three-phase current control strategies under charging conditions, we mitigated the torque risks during boost charging. This

research ultimately presents a highly competitive, low-cost, high-power, and torque-safe solution compatible with low-voltage DC charging stations for high-voltage EVs. The following Section I will introduce the evolution of the boost topology, analyze the advantages of the "3-port" scheme, and discuss design challenges. Section II will explain the control methods for reducing torque risks and charging current ripples. Section III will address motor magnet thermal risks and optimization strategies. Finally, an actual high-voltage electric drive motor sample and experimental validation demonstrate the effectiveness of the proposed solution.

Optimized Topology Solutions of Boost Charging

There is no need to discuss the mature DC-DC solution in integrated scheme as described in the introduction. The paper will start from the discussion of the integration solutions in electric drive motor of EVs. As mentioned in Introduction, few new EVs have been adopted boost topology configurations consisting of electric drive motor windings and inverter. However, to reduce current ripple and address motor heating issues, an obvious method is to add a power inductor to the electric drive motor. This is illustrated in [Figure 1](#), which shows a typical "4-port" integrated boost solution with an external power inductor. The additional boost circuit in this design includes a power inductor, two relays, a boost EMC filter, a boost capacitor, and two boost wires. The inclusion of an external power inductor requires extra vehicle space and also results in additional cost increases.

To avoid the additional external power inductor, article [15] proposes a method that reuses one phase of the motor's inductance as an external power inductor series with the other two phases of the motor along with the inverter to form a boost circuit. In this topology, the maximum charging current is limited by the one phase winding, resulting in relatively low charging power. On the other hand, the current in one phase of the motor is the sum of the currents in the two parallel-connected windings, so, there is a coupling relationship between the three phase-currents, which leads to a big challenge of the zero-torque control during boost charging. Another option as shown in [Figure 2](#), eliminating the external power inductor in the [Figure 1](#) scheme significantly reduces space occupancy and saves a lot of costs.

FIGURE 1 "4-port" Integrated Boost Topology with External Inductor.

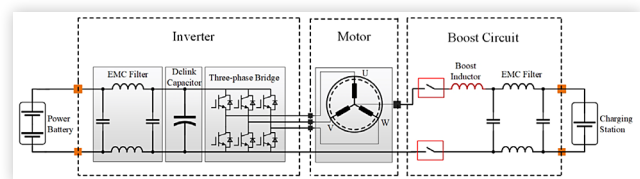
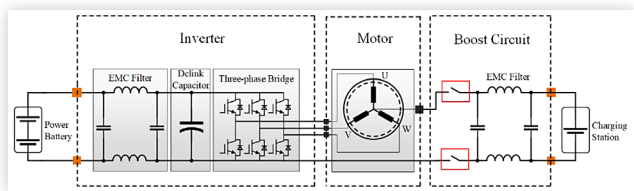
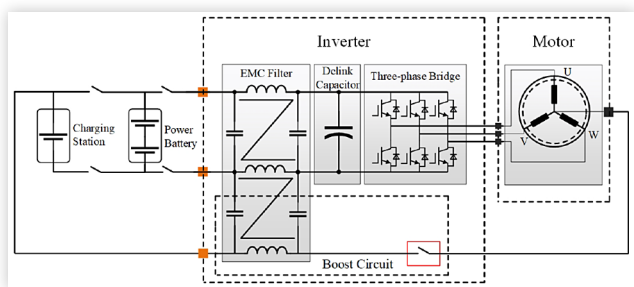


FIGURE 2 “4-port” Integrated Boost Topology.**FIGURE 3** “3-port” Integrated Boost Topology.

However, removing the external power inductor will cause the increase of the ripple in charging current, leading to the increase of motor losses, which needs the optimization of the motor design and control strategy. Additionally, similar to Figure 1, the boost input ports are independent from the bus side ports in this scheme, totaling four high-voltage ports, and the boost circuit still includes two relays and two boost wires.

In order to further improve the integration of the boost function with the electric drive motor and simplify the overall vehicle layout, this paper proposes a “3-port” highly integrated boost topology, as illustrated in Figure 3. This scheme eliminates the need for a dedicated negative wire for boost input by reuses the vehicle's DC-charging negative line. In the boost charging input circuit, only a single positive wire for boost is connected to the inverter, while sharing the EMC filter choke and capacitor packaging with the inverter input.

The proposed integrated solution does not require additional volume, significantly saving the overall vehicle layout space. Comparing with the structure of an inverter without a boost function, the new solution just needs one additional boost relay. This relay can be mounted on the motor or inverter housing where is suitable, thereby preserving the original motor/inverter design. Compared to the “4-port” approach, the “3-port” integrated boost solution described herein reduces the need for an EMC filter choke, one relay, and one boost cable, saving lots of vehicle costs.

In order to maximum the boost charging power, this solution fully employs the motor's three phase-windings capacity, allowing all three- phases to participate in boost charging. Without special modifications to the electric drive system, this configuration can increase charging power by 30% compared to a two phases windings parallel solution and by 60% compared to a one phase input series with the other two phases solution.

Moreover, vehicle application needs attention to torque control when using electric drive motor for boost charging conditions. Further discussion and analysis on zero-torque output control during boost charging conditions and the temperature rise of the motor magnets will be presented in the followings.

Control Algorithm and Performance of Boost Charging

When utilizing the motor's three phase-windings and inverter for boost charging while the vehicle is parked, it is essential to keep the torque control safe within the boost charging condition. It needs to prevent any unintended vehicle movement during the charging process to ensure the safety of both the vehicle and personnel. Additionally, to maintain a stable power supply from the DC-Charging station and to control the temperature of the boost circuit elements, it is important to optimize the charging current control strategy to minimize the ripple fluctuations so as the NVH performance in consideration. This section will analyze and solve these issues in detail.

Zero Torque Control of Motor

In boost mode, the output torque of motor is calculated as shown in Equation (1). The first part of the equation represents the electromagnetic torque generated by the interaction between the dq axes currents and the flux linkage. The second part is the torque generated by the zero-sequence current interacting with the third harmonic component of the permanent magnet flux linkage, which is relatively small. To ensure that the torque from the first part is zero, the i_q of motor must be controlled to equal zero.

$$T_e = \frac{3p_n}{2} \left[-L_d i_d i_q + (L_d i_d i_q + \psi_f i_q) - 6\psi_{3f} i_0 \sin(3\theta_e) \right] \quad (1)$$

Where:

- L_d and L_q are the direct and quadrature axis inductance; i_d and i_q are the direct and quadrature axis current,
- ψ_f and ψ_{3f} are the fundamental and third harmonic component of the permanent flux linkage,
- θ_e is the electrical position of the rotor; p_n is the number of pole pairs in the motor.

For different topologies, there are various methods for zero torque control. In the topology where one phase-winding is input and connected in series with the other two, the i_q can only be zero when the rotor position is at 0° if the currents in the parallel windings are equal. Within one cycle, this may result in a maximum output torque

exceeding 300 Nm, and considering the gearbox ratio, the wheel-end torque could exceed 2000 Nm, posing significant torque risks [15]. By analyzing different angles and adjusting the current distribution in the parallel windings, the range where i_q equal to 0 can be expanded to a specific 90° interval. However, outside this interval, the i_q current cannot be controlled to zero, and at certain angles, it may even reach 1.5 times the charging current.

For the topology where two phase-windings are in parallel, zero torque charging control can be achieved at all angles by using a combination of i_d and i_q to offset the torque produced by the zero-sequence current [16]. However, this method requires a fixed current ratio between the parallel windings, which limits the ability to charge both phases to their maximum current capacity within certain angular intervals, thereby restricting the maximum charging power.

The proposed deeply integrated three phase-windings parallel boost scheme employs a minimalist “3-port” topology. Compared to the topology where one phase-winding is input and connected in series with the other two, this approach allows for independent control of each phase-current, eliminating coupling issues and providing a natural advantage in zero torque control.

In this simplified “3-port” topology, a straightforward and reliable zero torque control strategy is to ensure three phase-currents equal, which results in both i_d and i_q being zero, thereby nullifying the electromagnetic torque component. The second torque component can be minimized during motor design by optimizing the third harmonic of the rotor flux linkage, satisfying the safety torque requirements under boost conditions.

Another engineering-friendly method involves automated calibration using a motor test bench. By employing torque sensors, the optimal current distribution for each rotor position can be determined to achieve zero output torque. Additionally, this method can utilize rotor and stator temperature estimating for optimization of the current ratios.

Comparing the two approaches, the first method is simple and reliable. By maintaining equal three phase-currents, the charging can occur at the maximum continuous current capacity of the windings at all angles, maximizing the utilization of the electric drive system's capabilities. In contrast, the second method depends on calibration and is subject to individual variations, requiring a sacrifice in the maximum charging power capacity due to current ratio constraints. Therefore, this paper opts for the first approach.

Phase-Interleave Control Strategy

This paper proposes a three phase-winding parallel boost charging scheme, comprising three independent boost circuits formed by the three phase windings and three bridge legs of the inverter, allowing for independent control of each phase current. As mentioned earlier, to achieve zero torque, this design employs a strategy where all three phase-currents are equal. In terms of carrier

phase control, there are two methods: in-phase and phase-interleave. The choice of carrier phase control strategy directly impacts current ripple during the boost charging process and has different requirements for input inductance. Therefore, selecting an appropriate strategy is crucial for minimizing current ripple and inductor value.

Closed-loop control with in-phase carriers is relatively simple but results in the addition of ripple currents from each phase. The total ripple current Δi_{charge} is derived as shown in Equation (2).

$$\Delta i_{charge} = \frac{3U_{out}T(D-D^2)}{L} \quad (2)$$

Where:

- U_{out} is the output voltage; L is the inductance,
- T is the switching period; D is the duty cycle.

Neglecting mutual inductance, phase-interleave control uses a 120° phase shift. As shown in Equation (3), the phase shifting allows the three-phase ripples to cancel each other out, theoretically bringing the ripple current to zero.

$$\Delta i_{charge} = \frac{U_{out}T(D-D^2)}{L} \left[\cos(0) + \cos\left(\frac{2}{3}\pi\right) + \cos\left(\frac{4}{3}\pi\right) \right] \quad (3)$$

As seen from Equations (2) and (3), compared to in-phase control, interleave control significantly reduces current ripple. However, the derivation above overlooks the influence of mutual inductance between windings and does not account for the specific current waveform under boost conditions. The following section provides a more precise calculation of ripple currents for both methods, considering the effects of mutual inductance and the actual boost current waveform.

According to the switching state of power devices, the phase-interleave method has six circuit modes. For example, in the first circuit mode, the power devices of the U phase and W phase are switched on, and the power devices of the V phase are switched off. So, in this mode, the voltage across the windings of U phase and W phase is charging input voltage, and the voltage across the winding of V phase is charging output voltage minus input voltage. These voltages are generated by the self-inductance of the windings and mutual inductance between the windings. The circuit mathematical model is derived as shown in Equation (4).

$$\begin{cases} U_{phaseU} = L_U \times \frac{di_U}{dt} + M_{UV} \times \frac{di_V}{dt} + M_{UW} \times \frac{di_W}{dt} = U_{in} \\ U_{phaseV} = M_{UV} \times \frac{di_U}{dt} + L_V \times \frac{di_V}{dt} + M_{VW} \times \frac{di_W}{dt} = U_{in} - U_{out} \\ U_{phaseW} = M_{UW} \times \frac{di_U}{dt} + M_{VW} \times \frac{di_V}{dt} + L_W \times \frac{di_W}{dt} = U_{in} \end{cases} \quad (4)$$

where:

- U_U, U_V, U_W are the voltage across the winding of U phase, V phase and W phase,
- L_U, L_V, L_W are the inductance of winding of U phase, V phase and W phase,
- M_{UV} is the mutual inductance between U phase and V phase, other similar symbols have the same meaning,
- i_U, i_V, i_W are the current flowing through the winding of U phase, V phase and W phase,
- U_{in} is the boost charging input voltage.

In other modes, the circuit model is constructed in a similar way. After calculating the current for each mode, the three phase-current and input charging current waveforms over two switching cycles are plotted, as shown in Figure 4 and Figure 5. Results show that under the phase-interleave control method, each phase-current consists of six segments due to mutual inductance, and the ripple

FIGURE 4 Phase current (Iphase U/V/W) and charge current (Icharge) under phase-interleave mode. Charge current is equal to the sum of the three phase currents.

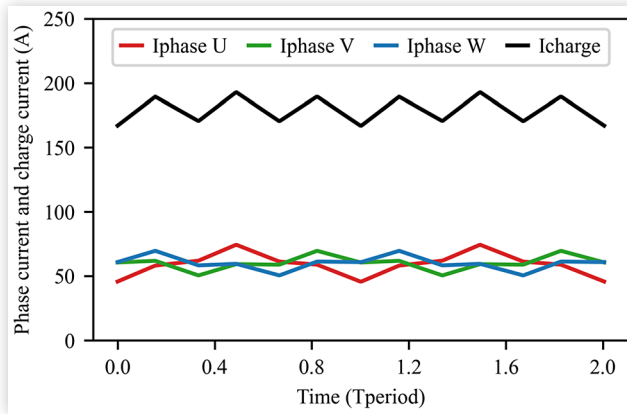
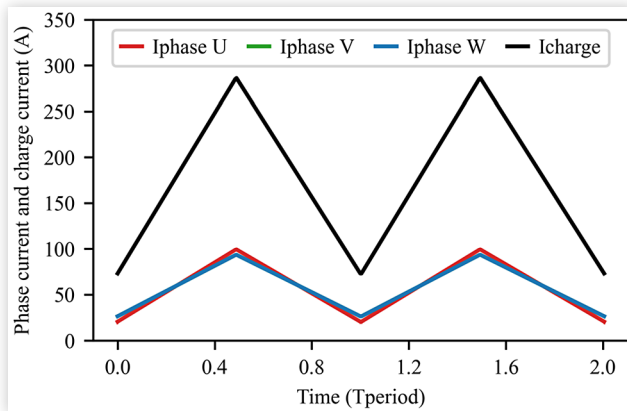


FIGURE 5 Phase current (Iphase U/V/W) and charge current (Icharge) under in-phase mode. The charging condition is consistent with that in Figure 4



amplitude of each phase-current is only one-fourth of that in the in-phase method. This is because, in the in-phase method, each phase-current change synchronously, reducing the effective inductance due to mutual inductance and thereby increasing the current ripple in each phase. Furthermore, the phase-interleave leads to the mutual cancellation of phase ripple currents, resulting in an input current ripple amplitude that is approximately one-ninth of the in-phase method.

Analysis of NVH Performance under Phase-Interleave Method

Under the boost mode, the implementation of phase-interleave control strategy would cause current ripple in the motor windings. The current ripple could lead to distortions in the airgap magnetic field, consequently generating electromagnetic force harmonics that induce electromagnetic vibration and noise. Therefore, it is necessary to analyze the electromagnetic force harmonics to assess the risk of NVH performance under the boost mode.

The electromagnetic vibration and noise are primarily caused by the radial electromagnetic force. According to the Maxwell stress tensor method, the radial electromagnetic force density can be calculated from the airgap magnetic flux density using Equation (5):

$$p_r = \frac{1}{\mu_0} (B_r^2 - B_t^2) \quad (5)$$

where:

- p_r is the radial electromagnetic force density,
- μ_0 is the vacuum permeability, and its value is $4\pi \times 10^{-7}$ H/m,
- B_r and B_t are the radial and tangential components of the airgap magnetic flux density, respectively.

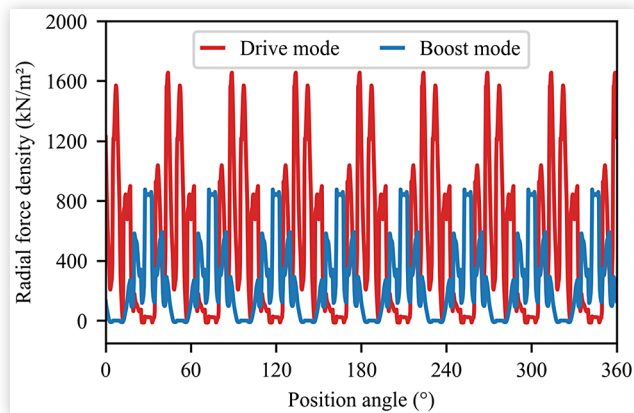
Figure 6 illustrates the comparison of electromagnetic forces of the motor under the boost mode and the normal drive mode. The comparison result presents that the amplitude of the electromagnetic force under the boost mode is significantly smaller than that under the drive mode. Therefore, the risk of vibration and noise under the boost mode is much lower than that under the drive mode. This conclusion will also be experimentally validated in experimental verification section.

Switching Frequency Strategy

The switching frequency of the motor inverter significantly affects the rotor eddy current loss, and the relationship can be described by Equation (6):

$$P_e = k_e B^2 f^2 V \quad (6)$$

FIGURE 6 Waveforms of radial electromagnetic force density under drive mode and boost mode.



where,

- P_e is the eddy current loss power,
- k_e is the eddy current loss correction factor,
- B and f are the amplitude and frequency of the magnetic field harmonic, respectively,
- V is the volume of the conductor.

On the one hand, increasing the motor inverter switching frequency could effectively reduce the amplitudes of current harmonics in the motor winding and the amplitudes of the high-frequency rotating magnetic field harmonics they generate, which has a positive effect on suppressing rotor eddy current loss.

On the other hand, as the switching frequency increases, the amplitude of eddy current induced in the rotor permanent magnets by high-frequency rotating magnetic field harmonics, which are generated by current temporal harmonics in the motor winding, also increases. Additionally, due to the skin effect, these high-frequency harmonics cause the eddy current to concentrate more towards the surface of the permanent magnets, reducing the conductive area and significantly increasing the eddy current density, thereby increasing the loss.

Therefore, it is necessary to consider both the above factors comprehensively to determine the optimal switching frequency that minimizes rotor eddy current loss.

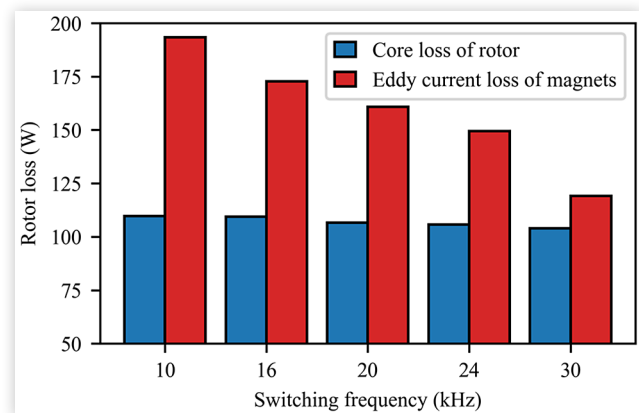
According to the simulation results in Figure 7, within the switching frequency range of 30 kHz, the comprehensive effect of increasing the switching frequency is to reduce the losses of the rotor.

Advanced Rotor Thermal Optimization

Reduction of Eddy Current Loss

Under the ideal driving condition, when the permanent magnet synchronous motor (PMSM) operates at the

FIGURE 7 Simulation results of rotor loss at different switching frequencies.



steady state, the fundamental component of the magnetic field generated by the stator rotates at synchronous speed and remains relatively stationary with the rotor, hence not inducing eddy current in the permanent magnets of the rotor. However, in the practical scenario, the non-sinusoidal distribution of the stator winding and the airgap permeance variation caused by stator slotting effect generate spatial harmonics. Additionally, inverter supply also induces significant temporal harmonics in the armature winding. Under the boost mode studied in this paper, the rotor remains stationary. Hence, stator armature reaction field harmonics rotate at high speed relative to the rotor under the rotor reference frame, exciting high-frequency eddy current in the permanent magnets of the rotor and significantly increasing eddy current loss.

Although rotor eddy current loss is smaller compared to stator iron loss and winding copper loss, the rotor has poorer heat dissipation condition than the stator. Under the boost mode, the motor is in a blocked state, and the amount of oil dumped by the rotor is significantly reduced, thereby further deteriorating the heat dissipation conditions of the rotor. Thus, even slight eddy current loss in the permanent magnets could lead to significant rotor temperature rise. Since the performance of permanent magnets is closely related to temperature, particularly for Neodymium (NdFeB) magnets commonly used in electric drives for new energy vehicles, their low Curie temperature, high conductivity, and large temperature coefficient are concerns. Excessive temperature not only degrades motor performance but can also cause irreversible demagnetization of the permanent magnets. Excessive temperature not only degrades motor performance but can also cause irreversible demagnetization of the permanent magnets. Furthermore, to meet the growing demand for high torque and power density, drive motors are developing towards higher speed, increasing the frequency of magnetic field harmonics and exacerbating rotor eddy current loss and temperature rise. Therefore, accurately calculating and mitigating rotor eddy current loss is critical to maintaining motor performance and reliability.

Segmenting permanent magnets is one of the most effective ways to reduce eddy current loss in permanent magnets. According to the equivalent magnetic circuit method, the relationship of eddy current loss before and after the permanent magnets are segmented along the z-direction and x-direction can be deduced in Equation (7).

$$\frac{W_{seg}}{W_{noseg}} = \begin{cases} \frac{\left[\frac{d(\Phi)}{N} \right]^2}{2(R_b + NR_a)} \approx \frac{(a+b)}{(Nb + N^2a)}, \text{ for } z\text{-direction} \\ \frac{\left[\frac{d(\Phi)}{N} \right]^2}{2(R_a + NR_b)} \approx \frac{(a+b)}{(Na + N^2b)}, \text{ for } x\text{-direction} \end{cases} \quad (7)$$

where:

- Φ is the magnetic flux of the magnetic circuit,
- a , b and d are the width, length, and thickness of the permanent magnets, respectively,
- N is the number of segments of the permanent magnets,
- R_a and R_b are the reluctance of the permanent magnet in the width and length directions, respectively.

From the above equation, it can be seen that the eddy current loss of the permanent magnet is significantly affected by the size of the permanent magnet. Therefore, by adopting an optimal permanent magnet segmentation method and adjusting the permanent magnet size to an appropriate value, the eddy current loss of the permanent magnet can be effectively suppressed.

To obtain the optimal segmentation method for permanent magnets, different segmentation methods are studied. As shown in Figure 8, four segmentation methods with varying numbers of pieces and directions are compared. The y-direction represents the magnetization direction of the permanent magnets, while the z-direction indicates the axial direction of the motor. Method I, method II and method III segment the permanent magnets into two, four and six pieces respectively along the z-direction, and method IV segments the magnets into four pieces along the x-direction. The distribution of eddy current paths in the magnets for each segmentation method is shown in Figure 9. It is evident that the segmentation of permanent magnets could

FIGURE 8 Four different segmentation methods of permanent magnets. (a) Method I: two segments along the z-direction. (b) Method II: four segments along the z-direction. (c) Method III: six segments along the z-direction. (d) Method IV: four segments along the x-direction.

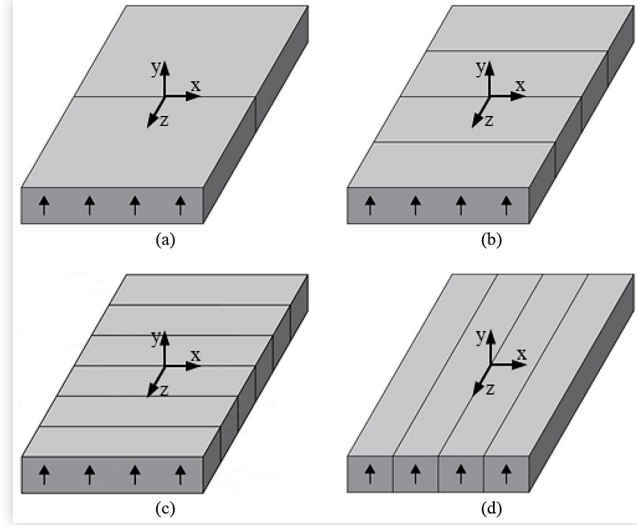
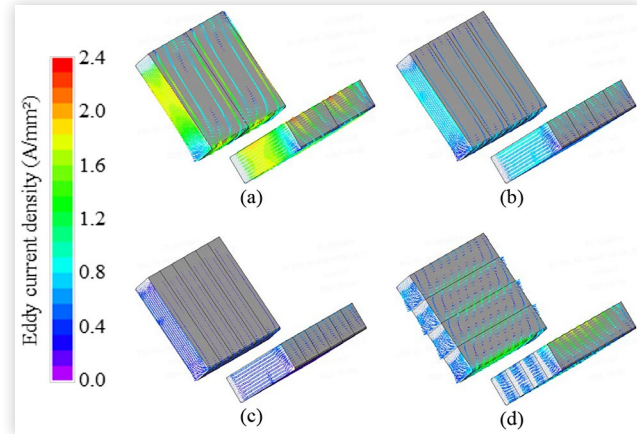


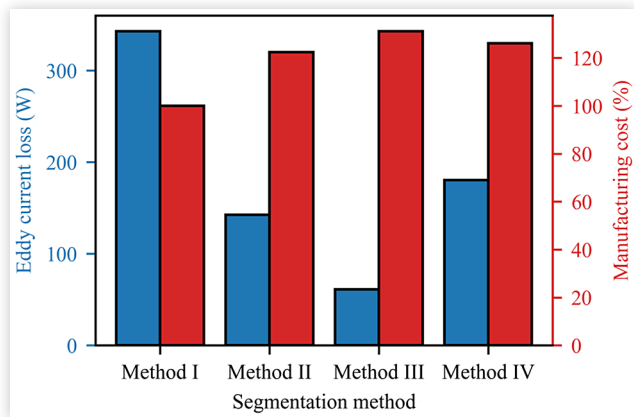
FIGURE 9 Eddy current distribution of four different segmentation methods. (a) Method I. (b) Method II. (c) Method III. (d) Method IV.



effectively isolate the eddy current paths and reduce the amplitude of eddy current density.

However, as different segmentation methods have different numbers of segments, insulation requirements, and insulation material consumption, resulting in different processing and manufacturing costs for permanent magnets, it is necessary to consider both the eddy current loss suppression effect and the processing and manufacturing cost to select the optimal segmentation method. Figure 10 compares the eddy current losses and manufacturing costs of four segmentation methods. As can be seen from the figure, the number of segments has a decisive influence on the eddy current loss, and by increasing the number of segments, the eddy current

FIGURE 10 Comparison of eddy current losses and manufacturing costs of four different segmentation methods.



loss can be reduced by more than 80%. Among the four segmentation methods, method III achieves the best suppression of eddy current loss. However, the excessive number of magnet pieces significantly increases the manufacturing cost. In contrast, method II can achieve an efficient eddy current loss suppression effect at a low-cost increase.

To further evaluate the demagnetization risk of permanent magnets under the boost mode, a computational fluid dynamics (CFD) simulation of the rotor temperature rise is essential. Figure 11 presents the simulated rotor temperature distribution with and without the segmented permanent magnets. The simulation conditions are specified as follows: a voltage boost from 440 V to 880 V at 40 kW, with cooling conditions set at an oil temperature of 30°C and an oil flow rate through the rotor of 3.14 liters per minute. As shown in Figure 11, the implementation of the optimized permanent magnet segmentation method results in a considerable reduction of the maximum rotor temperature rise by 28.4°C. The comparative analysis of rotor temperature rise confirms the effectiveness of the employed permanent magnet segmentation method. After segmentation, the temperature of the permanent magnets under the boost mode remains significantly below their maximum allowable operating temperature. In addition, under the boost mode, the current in the armature windings is relatively small, resulting in a modest demagnetizing field. As a result, the operating points of the permanent magnets remain above the knee point, thereby mitigating the risk of demagnetization.

Design of Cooling oil Path

In addition to adopting advanced permanent magnet segmentation method to suppress eddy current loss in permanent magnets from the source, the cooling oil paths have been optimized to further enhance the heat dissipation capability of the rotor and reduce the temperature rise. Figure 12 illustrates the distributions and shapes of the cooling oil holes on the rotor silicon sheet before and

FIGURE 11 Simulated temperature distribution of the rotor with and without segmented permanent magnets. (a) With segmented permanent magnets. (b) Without segmented permanent magnets.

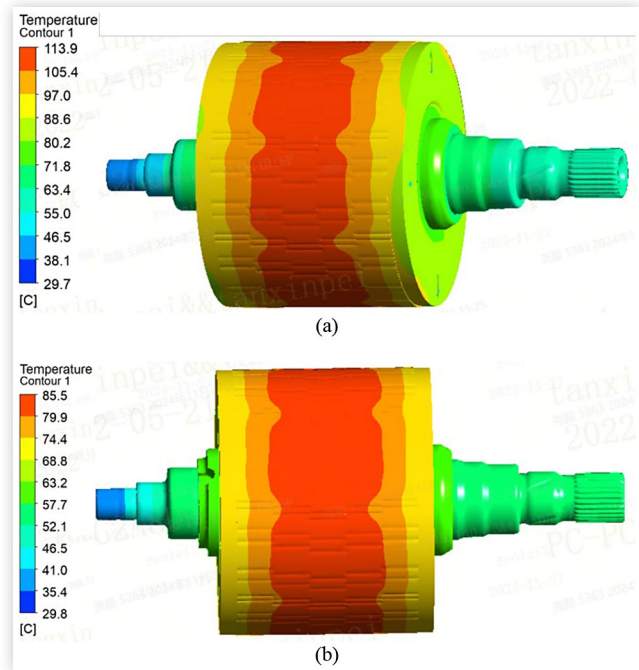
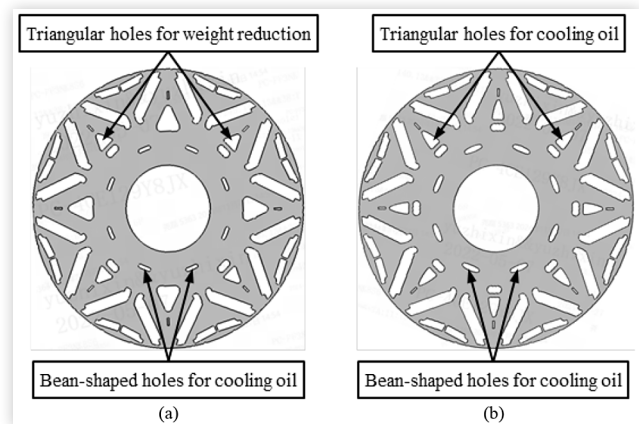


FIGURE 12 Distributions and shapes of holes for cooling oil before and after optimization. (a) Before optimization. (b) After optimization.



after optimization. Before optimization, eight bean-shaped cooling oil holes are evenly distributed circumferentially near the rotor shaft, with triangular weight-reduction holes positioned between the adjacent poles. After optimization, the triangular weight-reduction holes are replaced with the triangular cooling oil holes, and are connected in parallel with the bean-shaped cooling oil holes on the oil path. Besides, the shape of the bean-shaped cooling oil holes is also redesigned to ensure a more reasonable distribution of oil flow across the cooling oil branches when the rotor is skewed. Figure 13 shows the three-dimensional models of the rotor cooling oil

FIGURE 13 Paths of cooling oil before and after optimization. (a) Before optimization. (b) After optimization.

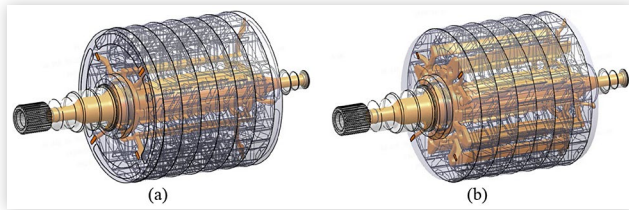
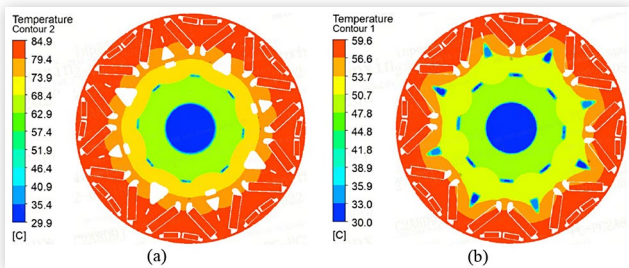


FIGURE 14 Simulated temperature distribution of rotor before and after cooling oil path optimization. (a) Before optimization. (b) After optimization.



paths before and after optimization, indicating a significant increase in the area of the cooling oil path from 0.018 m² to 0.051 m² by 186%, substantially enhancing the heat dissipation capability of the rotor.

Figure 14 shows the simulated temperature distribution of the rotor before and after the cooling oil path optimization, respectively. The simulation conditions are specified as follows: a voltage boost from 300 V to 600 V, with cooling conditions set at an oil temperature of 30°C and an oil flow rate through the rotor of 3.92 liters per minute. It can be seen from the simulation result in Figure 14 that after adopting the optimized cooling oil path design, the maximum temperature rise of the rotor is significantly reduced by 15.3°C, which is much lower than the maximum allowable operating temperature of the permanent magnets, avoiding the risk of irreversible demagnetization and improving the operation reliability and safety of the motor.

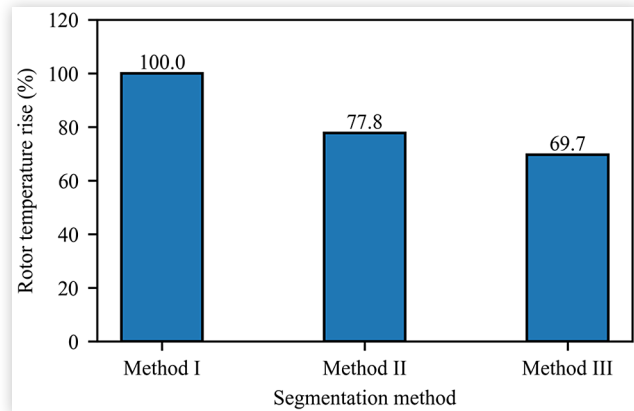
Experimental Verification

Verification of Motor Temperature Optimization Method

Effect of Permanent Magnets Segmentation In order to verify the effect of segmenting permanent magnets on rotor temperature rise and select the optimal segmenting method, different segmenting methods were tested.

During the test, two, four, and six segments along the z-direction were verified respectively on the same

FIGURE 15 Rotor temperature rise of three different segmentation methods. Method I for two segments, Method II for four segments, Method III for six segments.



motor. As shown in Figure 15, segmenting permanent magnets has a highly significant reduction effect on rotor temperature rise. Specifically: from two segments to four segments, the rotor temperature rise is reduced by 22.2%; from four segments to six segments, the rotor temperature rise is reduced by 8.1%.

Effect of Switching Frequency Increasing To verify the effect of rotor-temperature rise after increasing the switching frequency, the rotor temperature rise under different switching frequencies was tested on the same motor. The switching frequencies of 16kHz, 20kHz, 24kHz, and 30kHz were verified respectively.

As shown in Figure 16, as the switching frequency increases from 16kHz to 30kHz, the rotor temperature rise continues to decrease, and it is reduced by 17.0% at 30kHz compared with 16kHz.

Effect of Oil Path Optimization Rotors with different oil cooling paths were tested on a motor to verify the effect of rotor temperature rise after optimization measures.

From the test results, the rotor temperature is significantly reduced after cooling oil path optimization. Under

FIGURE 16 Rotor temperature rise at different switching frequency, from 16kHz to 30kHz.

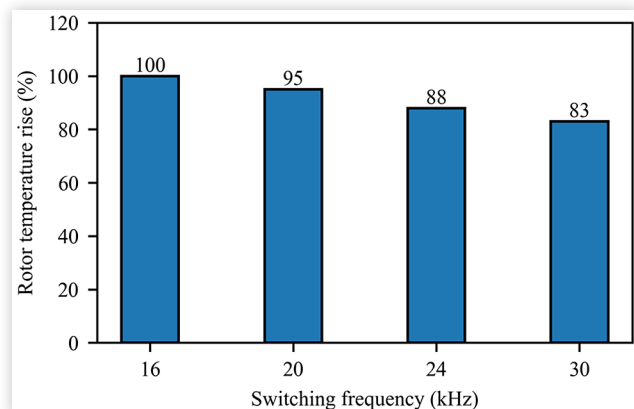
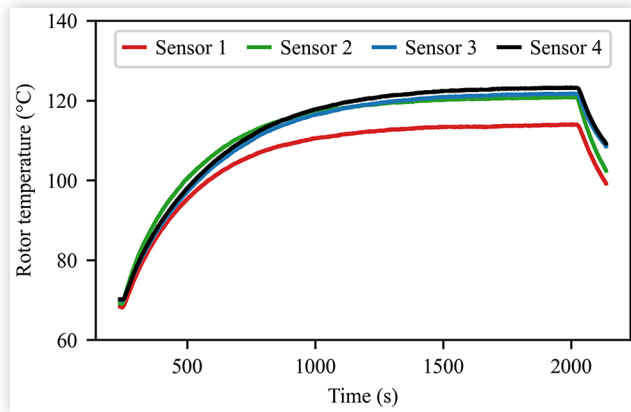


FIGURE 17 Permanent magnets temperature at the middle position of the rotor. Sensor1 to Sensor 4 measured the temperature of all 4 types of permanent magnets used in this motor.



the same working condition, the rotor temperature rise can be reduced by 10.2%, compared with the original scheme.

Verification of Rotor Temperature on Motor Assembly Based on the verification results above, optimization measures like dividing the magnet into four segmentations, using a 30kHz switching frequency, and optimizing the oil cooling path are adopted.

The rotor temperature verification under the worst booster charging condition was conducted on the electric drive assembly. The permanent magnet temperature at the middle position of the rotor was continuously detected, as shown in the Figure 17.

It can be seen from Figure 17, after continuous boost charging, the rotor temperature tends to be stable, the rotor temperature change within 10 minutes is close to 1°C, and the maximum rotor temperature is about 123.2°C, which is close to the simulation analysis result, and has a

large margin in line with the rotor temperature design requirements.

Verification of Sound Pressure Level

To evaluate the impact of boost mode on motor NVH performance, the sound pressure level of the motor under the boost mode were tested and compared with that under the drive mode at various speeds, as shown in Figure 18. The comparison result indicates that the sound pressure level of the motor under the boost mode is approximately 50 dBA, significantly lower than that under the drive mode, which is up to 71 dBA at high speed condition. Thus, the boost mode will not increase the risk of vibration and noise in the motor.

Verification of Ripple Current and Capacitor Temperature Rise

The ripple current of the input capacitor was tested on the assembly under the worst boost charging condition.

Measured results in Figure 19 show that the effective value of the input boost charging input current is 5.10A under the worst working conditions, which meets the design capability of capacitor current ripple. Furthermore, the temperature of the boost capacitor is lower than 105°C, which meets the temperature design requirements.

Verification of Torque Output

The output torque under boost charging mode is verified on the assembly at multiple angles, and the charging current is controlled at the maximum value. The measured output torque of the motor is about 2.5Nm.

FIGURE 18 Measurement results of sound pressure levels of motor under drive mode and boost mode. (a) Drive mode. (b) Boost mode.

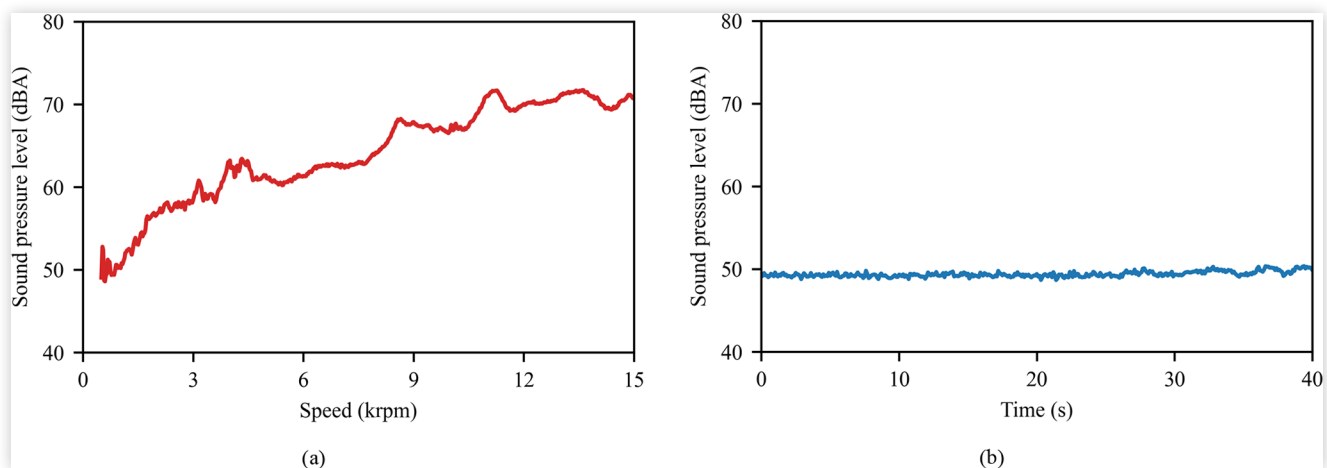
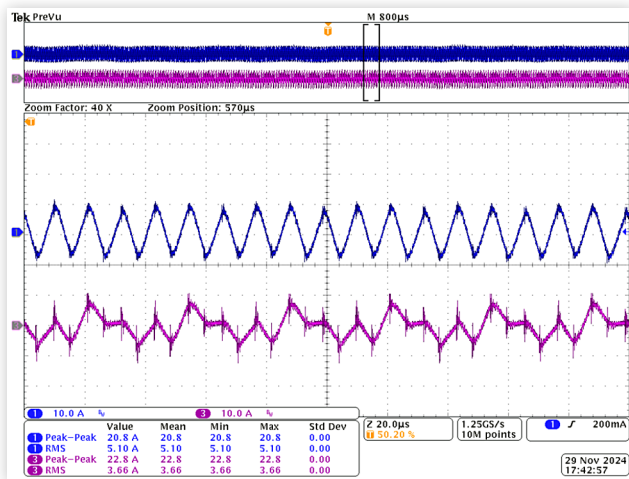


FIGURE 19 Ripple current test result. The top curve represents the ripple current of the capacitor at the input circuit of boost charging, and the bottom curve represents the ripple current of the phase current.



Converted to the output torque of the shaft end, it can meet the torque requirements. When the vehicle is parked on a ramp and the IPB on one side fails, the torque generated will not cause unintended vehicle movement, even at maximum charging current.

Summary/Conclusions

With the rapid market penetration of high-voltage EVs, extensive research and application have been conducted on charging solutions compatible with lower voltage DC-Charging stations. The approach of utilizing electric drive motor for boosting charging has gradually become a predominant direction. Building upon prior research, this paper focuses on practical vehicle applications and product design aspects. Considering the minimal integration cost for boost charging functionality, we designed a minimalist “3-port” scheme that significantly reduces the increase in components required for compatible boost charging while fully leveraging the current capacity of the three phase-windings. Through control strategies, torque safety during boost charging conditions is achieved without restricting the motor's angular position when parked, ensuring robust software control strategy performance. It addressed the issues of magnet steel eddy current losses and heating by employing segmented magnet designs and rotor oil cooling enhancements, systematically implementing a highly competitive vehicle engineering solution.

Real prototype integration and experimental validations of the boost charging design in vehicle electric drive system have proven the effectiveness of segmented magnets design, voltage control strategies, and enhanced cooling solutions. Based on a 250kW 800V permanent magnet drive motor, it achieved a maximum boost charging power of 180kW and a boost charging efficiency

of 98%, with no risks of motor overheating or torque safety concerns.

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